assignment 13 distance ladder

This is a preview of the draft version of the quiz

Started: Apr 29 at 8:11pm

Quiz Instructions

2 attempts

Question 1 4.2 pts

In astronomy, a *standard candle* is an object with a known

apparent brightness.

distance.

luminosity.

surface temperature.

Question 2 4.2 pts

The distance Earth-Sun is about 150

thousand km

million km

million m

billion km

Question 3 4.2 pts

All the rays from the Sun reaching the Earth are parallel to each other (T or F)

True

The Sun rays don't reach the EArth

False

Question 4 4.2 pts

In the 60s, the actual known distance Sun-Earth (1AU) was found using

- transit of a planet in front of the Sun
- spectroscopy
- a radar
- the parallax method

Question 5 4.2 pts

What is the most accurate way to determine the distance to a relatively nearby star (within the Milky Way. With a distance smaller than 100 l.y away from us (using the EArth and using hipparcos)?

- radar ranging
- applying Hubble's law
- using white dwarf supernovae
- stellar parallax
- using Cepheid variables

Question 6 4.2 pts

Who measured the distance to the Moon using a solar eclipse and a lunar eclipse?

- Erasthotenes
- Hipparchus
- Copernicus
- Aristarchus

Question 7 4.2 pts

A star with a large parallax

- is not moving with respect to Earth.

is moving at a great speed with respect to Earth.

is moving at a slow speed with respect to Earth.

is at a short distance from Earth.

is at a great distance from Earth.

Question 8 4.2 pts

Which planet is at a distance of about 70% of distance Earth-Sun ? see slide

Venus

Mars

Mercury

Jupiter

Question 9 4.2 pts

An absolute value for the distance Sun-Earth (1AU) was found using the transit of
(observed by James Cook in 1769 in Tahiti).

Mercury

Mars

Venus

Moon

Question 10 4.2 pts

suppose you measure the parallax angle for a particular star to be 0.1 arcseconds . What is the distance to the star (equation - slide12)

10 light years

0.1 parsecs

10 parsecs

0.1 light years

Question 11 4.2 pts

building up on previous question. 10 pcs is about

50 light years

1 light year

30 l.ight years

10 light years

Question 12 4.2 pts

In cass we discussed how Eratosthenes : "measured " the size of the Earth using a stick and basic geometry. To get the rays of the Sun just perpendicular to the ground and reaching the bottom of a well, the experiment has to be done during

spring equinox

the summer solstice

fall equinox

winter solstice

Question 13 4.2 pts

The circumference of the EArth is about ___________ times the distance between Alexandria and Syene

see the lecture about:Eratosthenes

5000

50

100

10

Question 14 4.2 pts

If a star is found by spectroscopic observations to be about 500 parsecs distant, its parallax is: (slide 13)

- 0.02"
- 0.002"
- 0.5"
- 0.2"

Question 15 4.2 pts

Hipparcos was a space probe that measured distances to stars with a max distance of about

(in orbit around the Earth)

- 100 l.y
- 10 l.y.
- 30 l.y.
- 50 l.y.

Question 16 4.2 pts

The distance Earth-Jupiter is about

- 5 A.U.
- 10A.U.
- 3 A.U.
- 1.5 A.U

Question 17 4.2 pts

Copernicus measured the relative distance between the planets and the Earth.

The green dot is the EArth. The Yellow dot is the Sun. The cross shows the location of (slide 7)

- Venus
- Mars
- Jupiter.
- Mercury

Question 18 4.2 pts

What are the steps of the distance ladder

- radar, HR diagram, parallax, cepheids, rotation galaxies, supernovae type IA
- radar, parallax, HR diagram, cepheids, rotation galaxies, supernovae type IA
- HR diagram, radar, parallax, cepheids, supernovae type IA rotation galaxies
- radar, cepheids, parallax, HR diagram, rotation galaxies, supernovae type IA

Question 19 4.2 pts

If the distance from us to a star is multiplied by 2, its parallax is

- divided by 4
- multiplied by 4
- multiplied by 2
- divided by2

Question 20 4.2 pts

If astar #1 is at 300 light years away from us and start #2 is at 100 light years away from us. What is true?

Remember the inverse square law. (square!)

- star2 is 3 times brighter
- star2 is 4 times brighter
- star2 is 9 times brighter
- star2 is 2 times brighter

Question 21 4.2 pts

Cluster A is closer from our Sun than Cluster B. The clusters are groups of stars gravitationally bound in other galaxies. The stars of each clusters are plotted on a HR diagram. What is true

- The stars will be randomly distributed in a HR diagram. There is no pattern
- the main sequence of cluster B is shifted vertically down relative to A
- The main sequences of the clusters are surimposed,
- the main sequence of cluster A is shifted vertically down relative to B

Question 22 4.2 pts

A cluster of stars is observed and the stars are plotted in a HR diagram. No O, B, A, F or G stars are observed. What is true (see slide/lecture)

- The cluster is younger than 1 billion years
- We can not determine the age of a cluster.
- The cluster is at least 10 billion years
- The cluster is younger than 10 million years

Question 23 4.2 pts

The cluster the Hyades is 7.5 brighter than the Pleiades. That means

(slides and lecture)

- I spaced out in class
- The pleiades are 2.5 times closer
- The Pleiades are 7.5 times farther away
- The PLeiades are 2.5 times farther away

Question 24 4.2 pts

so if the Hyades is 50 pc away from us (parallax method determined that) then the pleiades is

- 200 pc away
- 125 pc away
- whaaaat
- 375 pc away

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